

Writing Workshop #1: Finding the Right Word

“It is said that back in the 1940s, the following message was prominently displayed at the front of the main chemistry lecture hall at a major university: ‘The English language is your most versatile scientific instrument. Learn to use it with precision.’”*

The American author and journalist Samuel Langhorne Clemens (Mark Twain) put it more colorfully: “The difference between the right word and the *almost*-right word is the difference between *lightning* and *lightning bug*.”†

Many words in English have the same denotation (the literal dictionary definition) but subtly different connotations (additional shades of meaning that have cultural or emotional associations beyond the literal definition). For example, I seek to write *precisely*. Does that make me *painstaking* and *meticulous* or *finicky* and *nit-picking*? The four adjectives used in the previous sentence all have the denotation of “aspiring to precision,” but their connotations are different. The first two indicate that I’m a careful practitioner of my craft; the latter two imply that I’m a fussy pain-in-the-ass. (Which description actually applies to me will be revealed as the semester progresses.)

Choosing the right word, the one having the proper connotation to express your intended meaning exactly, requires having an extensive vocabulary. In science writing, using words precisely is essential to properly and unambiguously convey‡ subtle differences and abstract concepts.

Knowing the right word is also essential for conciseness—another hallmark of good scientific writing. For example, what if you were trying to write about this object



and you didn’t know the word *key*? You’d have to resort to calling it “a small, flattened metal object, consisting of a head and a stem, that is used for opening and closing mechanical locks. The thin stem, which is grooved, has a series of indentations along one or both sides. The stem is inserted into the cylinder of the lock, such that the grooves and indentations line up with the tumblers of the lock. A twisting motion to the head manipulates the tumblers to open or close the lock.” Neither concise nor immediately understandable—the twin curses of bad scientific writing. pp

* G.L. Trigg, “Editorial,” *Phys. Rev. Lett.* **42**, 747–748 (1979).

† Mark Twain, *The Wit and Wisdom of Mark Twain* (Dover Publications, 1998).

‡ Yes, I split an infinitive. Twice. Seventh-grade spinster English teachers all over America are having attacks of the vapors. Get over it. There is no “rule” against splitting infinitives. Split as many as you want. But do so thoughtfully and deliberately, not out of sloppiness.

One of the best ways to expand your vocabulary is to read a lot—and not just scientific papers and textbooks. Read novels and magazines and cereal boxes and newspapers. Keep a dictionary[§] next to your reading spot, and when you encounter a word that you don't know, **look it up** and put a check mark in the margin next to the word. When a word accumulates five check marks, resolve to *learn* the word and commit it to your vocabulary. Knowing lots of words will help you choose exactly the right one to express your meaning precisely.

For this exercise, write down as many alternatives as you can for the underlined words in the following eight sentences. These words are overused in scientific writing, are used incorrectly in the example sentence, or are wimpy, meaningless adjectives that proliferate in sloppy technical writing.

Try to think of at least **five words** that are more precise or descriptive for each underlined word, but don't use slang and don't use words that you would not expect to see in a scientific article. For example, don't use *ravishing* or *voluptuous* or *hot* for *beautiful* in Sentence 8. In some cases, you may decide that *no* adjective is better than the original word. (In good science writing, we write with exact nouns and verbs, not adjectives and adverbs.)

DO NOT cheat and use Word's built-in or an online thesaurus for this exercise. Instead, use the practice as a diagnostic to assess your own vocabulary database. Your grade will not be adversely affected if you cannot come up with five words, but do your best. Upload your exercise to the assignment portal on the *my.physics.illinois.edu* at the end of today's workshop.

All examples are taken from papers published in *Physical Review Letters*. Bibliographic citations have been omitted to protect the guilty. Additional commentary on each sentence by Ms. Particular is noted in the posted solutions.

1. Experiments suggest that the celebrated “boson peak”—a low-frequency vibrational feature characteristic of amorphous materials—may be related to a well-known phonon singularity in ordered crystalline materials.

- Characteristic
- Well-understood

You can also remove this word, since the word is largely opinionated.

2. Support of organizations like AP over time is key to ensuring the project's long-term success. (Brownie points for spotting and fixing the four-letter word (FLW); FLWs are discussed at <http://people.physics.illinois.edu/Celia/MsP/Four-LetterWords.pdf>.)

- Critical

[§] Merriam-Webster's *Collegiate Dictionary* and *The Oxford Dictionary for Scientific Writers and Editors* are recommended.

Upload your exercise to gradescope at the end of today's workshop.

- Necessary
 - Required
3. The physical system we introduce appears as an interesting model system in the context of complex systems.

Rich, complex, relevant

4. The control of vortex configurations in nanoferroelectrics would open exciting opportunities for designing revolutionary devices.
- 1. Different, unique, interesting
 - 2. Novel, new,
5. Proposals for long-distance quantum communication rely on the entanglement of matter-based quantum nodes through optical communications channels, but the entangling light pulses have poor temporal behavior in current experiments.

Subpar, bad, suboptimal, unfavorable, low quality

6. Strong electronic correlations in complex oxides lead to a wealth of electronic properties and rich phase diagrams.

Various, trove, myriad

7. The principle of frequency doubling became fully comprehensible with subsequent theoretical analysis; the rapidly ensuing, parallel development of experiment and theory essentially marked the birth of the diverse field we now call nonlinear optics.

Remove this word, the word isn't doing anything.

8. The key to achieving some beautiful results has been the ability of the Duke University team to engineer a waveguide that will exhibit strikingly different dielectric and magnetic permeability properties, at appropriate fundamental and harmonic frequencies, so as to elicit negative refraction for the optical harmonic.
- 1. Remarkable, interesting,
 - 2.

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If this exercise was difficult because your vocabulary isn't as expansive as it could be, invest in a second-hand copy of J.J. Rodale's *The Synonym Finder*, eds. Laurence Urdand and Nancy LaRoche (Rodale Press, Emmaus, PA, 1978), and **use** it. *Rodale* is one of the professional writer's best friends and constant companions.

A more deliberate approach to increasing your vocabulary is presented in Norman Lewis's *Word Power Made Easy*. Lewis associates words in logical groups—useful for learning the nuances of similar words—and provides plenty of hands-on practice materials and worksheets. Best of all, his comprehensive book costs less than \$10.

A fun way to expand your vocabulary (*and* prepare for the GRE, **and** contribute to the fight against world hunger—all at the same time) is to go to [Freerice - Home](#) and practice.

If you score higher than 2750 send us a screen shot for bragging rights in PHYS 496.

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